

Q.3 Explain the working principle of transformer?

Ans:- Principle of Operation :- When alternating voltage V is applied to the primary winding of a transformer a current (termed as exciting current, I_0) flows through it as shown in Figure 1. The exciting current produces an alternating flux (ϕ) in the core, which links with both the winding (primary and secondary). According to Faraday's law of electromagnetic induction, the flux will cause self-induced emf E in the primary and mutually induced emf will oppose the applied voltage and in magnitude this primary induced emf is (almost) equal to the applied. Therefore, in brief we can say emf induced in the primary winding is equal and opposite to the applied voltage.

When a load is connected to the secondary side, current will start flowing in the secondary winding. Voltage induced in the secondary winding is responsible to deliver power to the load connected to it. In this way power is transformed from one circuit (primary) to another (secondary) winding through a magnetic circuit by electromagnetic induction. This is the working principle of the transformer. The induced emf in the secondary E_2 is also in phase opposition to the applied voltage V_1 at primary. If the secondary is open circuited, terminal voltage V_2 at the secondary is equal in magnitude and in phase with the induced emf at secondary.

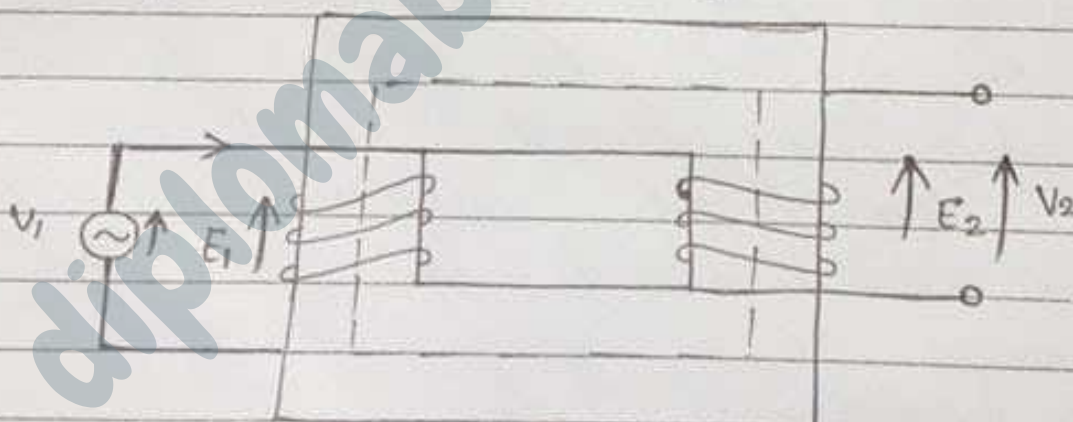
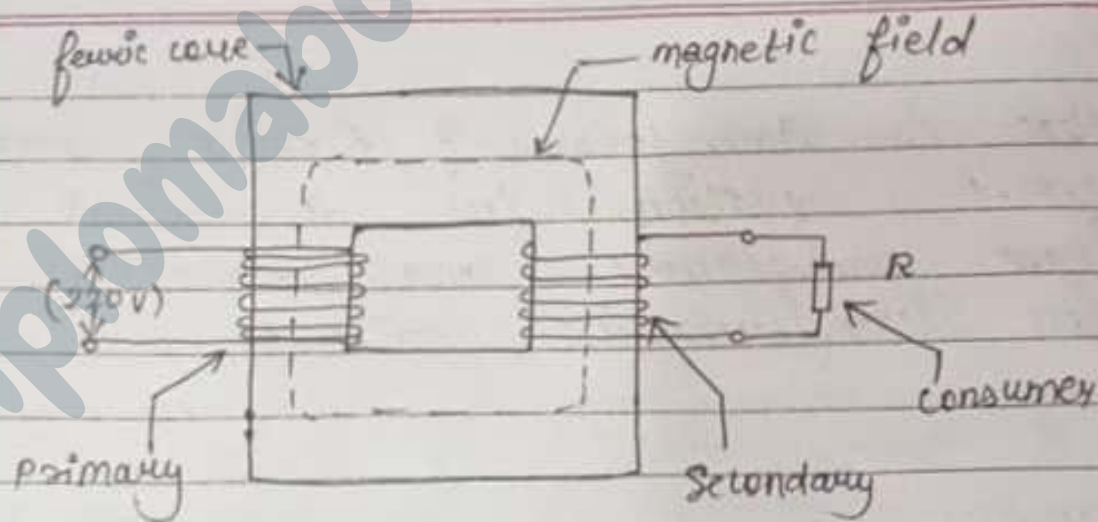


Fig:1 Schematic diagram of single phase transformer.

Q4. What is Transformer? Derive and e.m.f equation for a single phase transformer and deduce expression for Transformation ratio?

Ans:- A Transformer is a device which transfers electric current from one circuit to another, usually by the principle of mutual induction. During this process, the frequency remains constant whereas the voltage can be increased or decreased according to the need.

This transfer of electricity occurs with the help of two coils. One of which is known as the primary coil, which is connected to a source of alternating current. The other is known as the secondary coil, and it is connected to an external circuit. This constitutes the basic structure of the transformer and is shown below:



EMF equation of the Transformer:-

Consider that an alternating voltage V_1 of frequency f is applied to the primary as shown in Fig (a). The sinusoidal flux ϕ produced by the primary can be represented as:

$$\phi = \phi_m \sin \omega t$$

- The instantaneous e.m.f. e_1 induced in the primary is

$$e_1 = -N_1 \frac{d\phi}{dt} = -N_1 \frac{d}{dt} (\phi_m \sin \omega t)$$

$$= -\omega N_1 \phi_m \cos \omega t$$

$$= -2\pi f N_1 \phi_m \cos \omega t$$

$$= 2\pi f N_1 \phi_m \sin(\omega t - 90^\circ)$$

- In a pure inductive circuit lags behind the voltage by 90° .

- Since there are no iron losses, flux ϕ is in phase with i_m .

- It is clear from the above equation that maximum value of induced e.m.f in the primary is :-

$$E_{m1} = 2\pi f N_1 \phi_m$$

- The r.m.s value E_1 of the primary e.m.f is

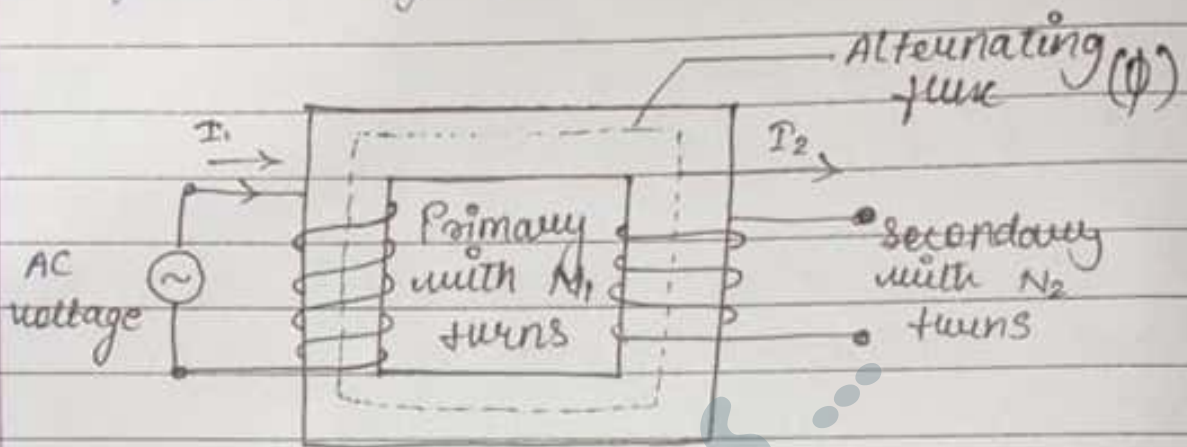
$$E_1 = \frac{E_{m1}}{\sqrt{2}} = \frac{2\pi f N_1 \phi_m}{\sqrt{2}}$$

$$\text{or } E_1 = 4.44 f N_1 \phi_m$$

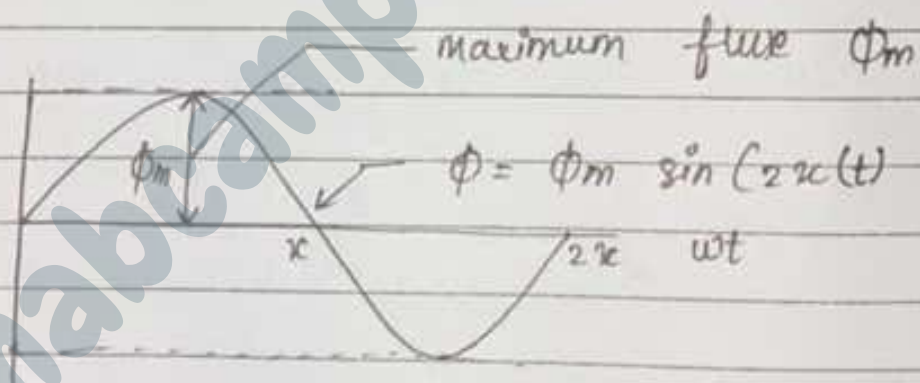
$$E_2 = 4.44 f N_2 \phi_m$$

- In an ideal transformer, $E_1 = V_1$ & $E_2 = V_2$

Note :- It is clear from exp. (i) above that e.m.f. E_1 induced in the primary lags behind the flux ϕ by 90° . Likewise, e.m.f. E_2 induced in the secondary lags behind flux ϕ by 90° .



(a) Elementary transformer.



(b) Variation of the flux with time.

Voltage Transformation Ratio (k)
As defined above,

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} = k$$

Where $k = \text{constant}$

This constant k is known as voltage transformation ratio.

- (a) If $N_2 > N_1$, i.e. $k > 1$, then the transformer is called step-up transformer.
- (b) If $N_2 < N_1$, i.e. $k < 1$, then the transformer is called step-down transformer.

Q5. What is transformer ratio or turn ratio? What is its significance?

Ans: Transformer ratio (k):

The ratio of secondary voltage to the primary voltage is known as transformation ratio.

OR

The ratio of secondary winding induced voltage to the primary winding induced voltage is known as transformation ratio.

$$K = \frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

Significance :-

If the secondary windings has half the number of turns of the primary winding, then the secondary voltage will be half the voltage across the primary winding likewise, if the secondary winding has twice the number of turns of the primary winding, the secondary voltage will be double the primary voltage.